

Course

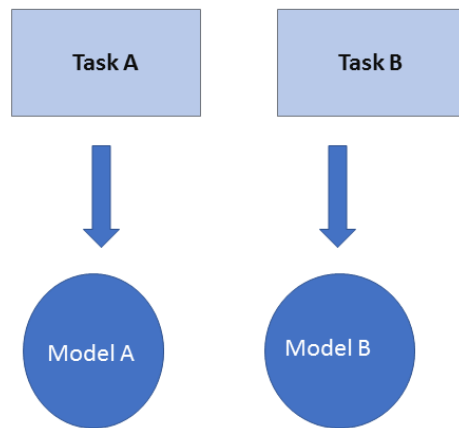
« *Deep Learning* »

Sid-Ahmed Berrani

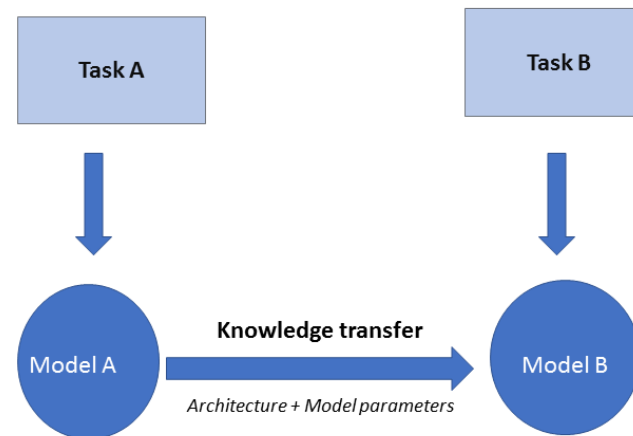
ENSIA
2024 – 2025

Transfer Learning

Transfer Learning is a technique that involves using a model trained on one dataset to perform a specific task as a starting point for another, similar task.



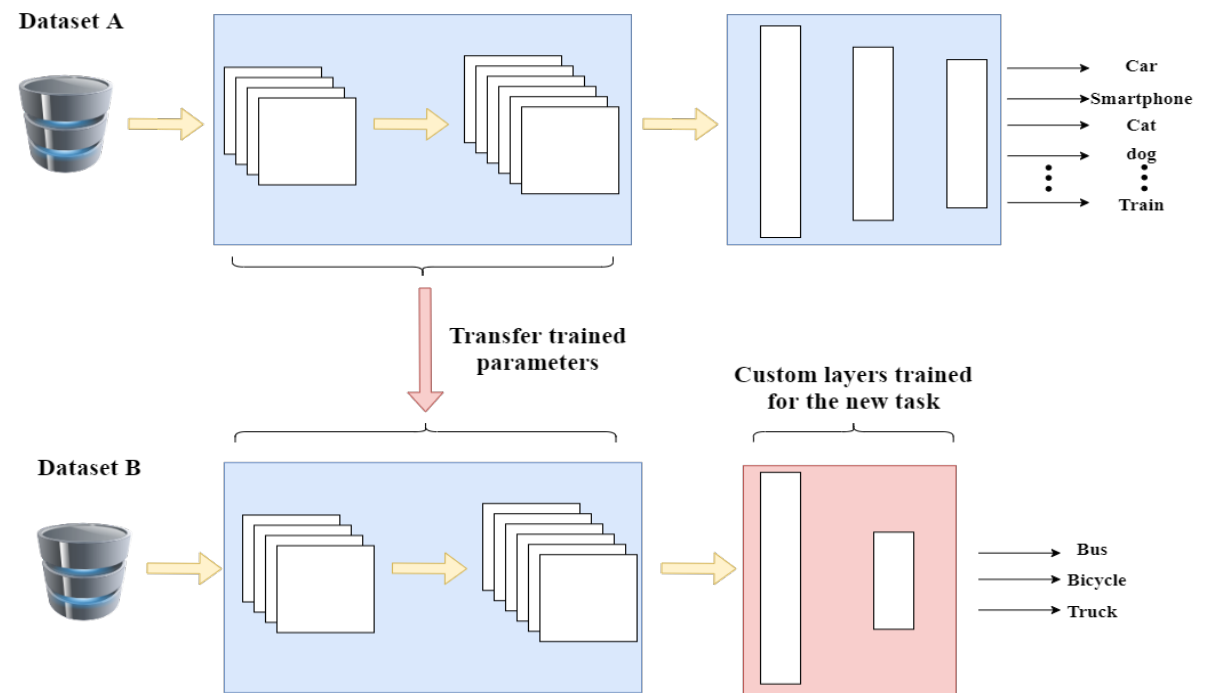
Training from scratch



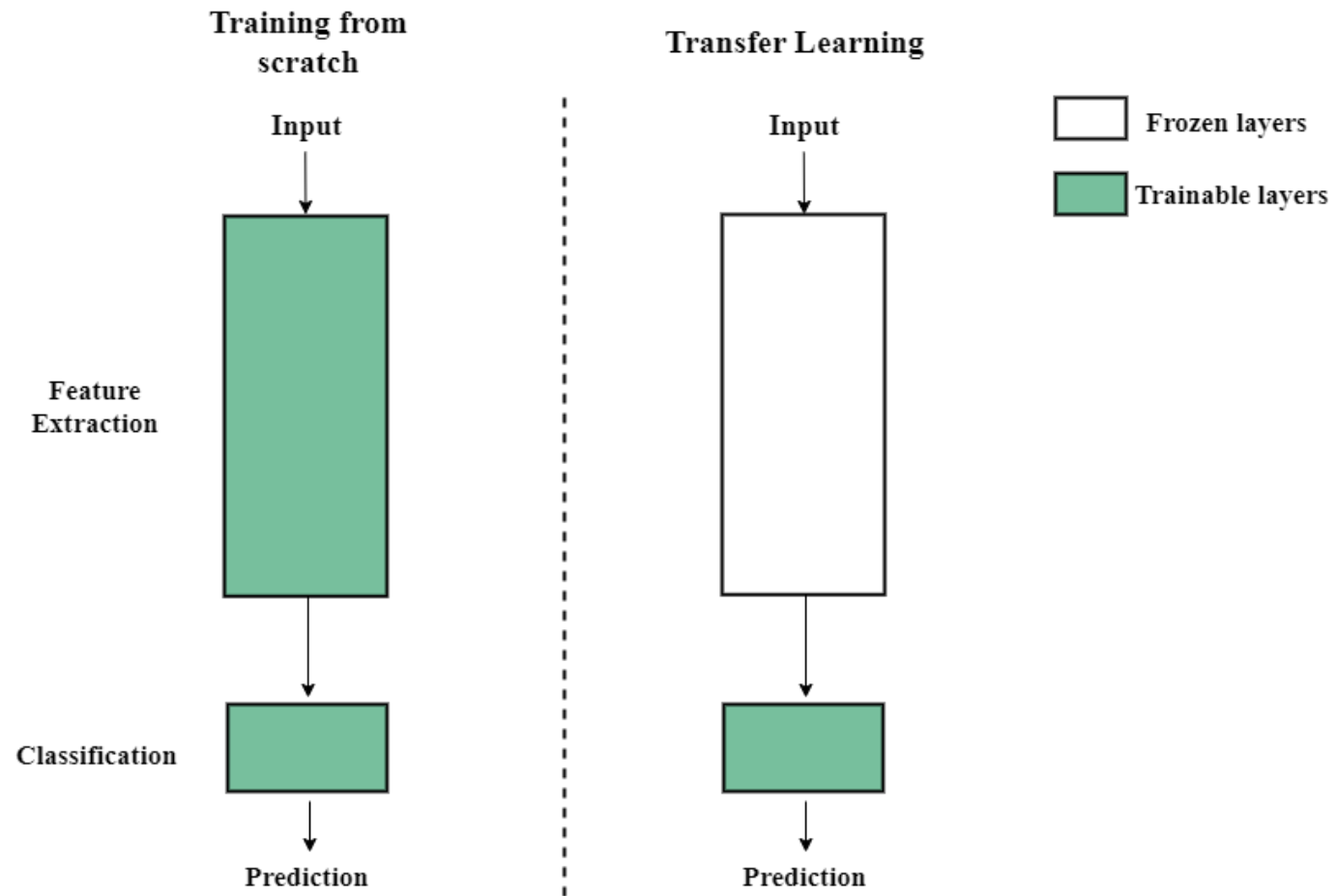
Transfer Learning

Transfer Learning Procedure

- Leverage the knowledge acquired by the pre-trained model by reusing the initial layers responsible for feature extraction in the new model.
- Customize the final layers used for classification by replacing them with other layers that are better suited to the new task (e.g., number of classes, etc.).



Transfer Learning Procedure



Why Transfer Learning?

Generally, training a neural network model requires a large amount of data, as well as a highly powerful platform to handle the intense computational demands of learning.

With Transfer Learning, a model that has already been trained on a wide variety of data can be adapted to a new task using relatively little data because it has acquired sufficient knowledge to adjust accordingly.

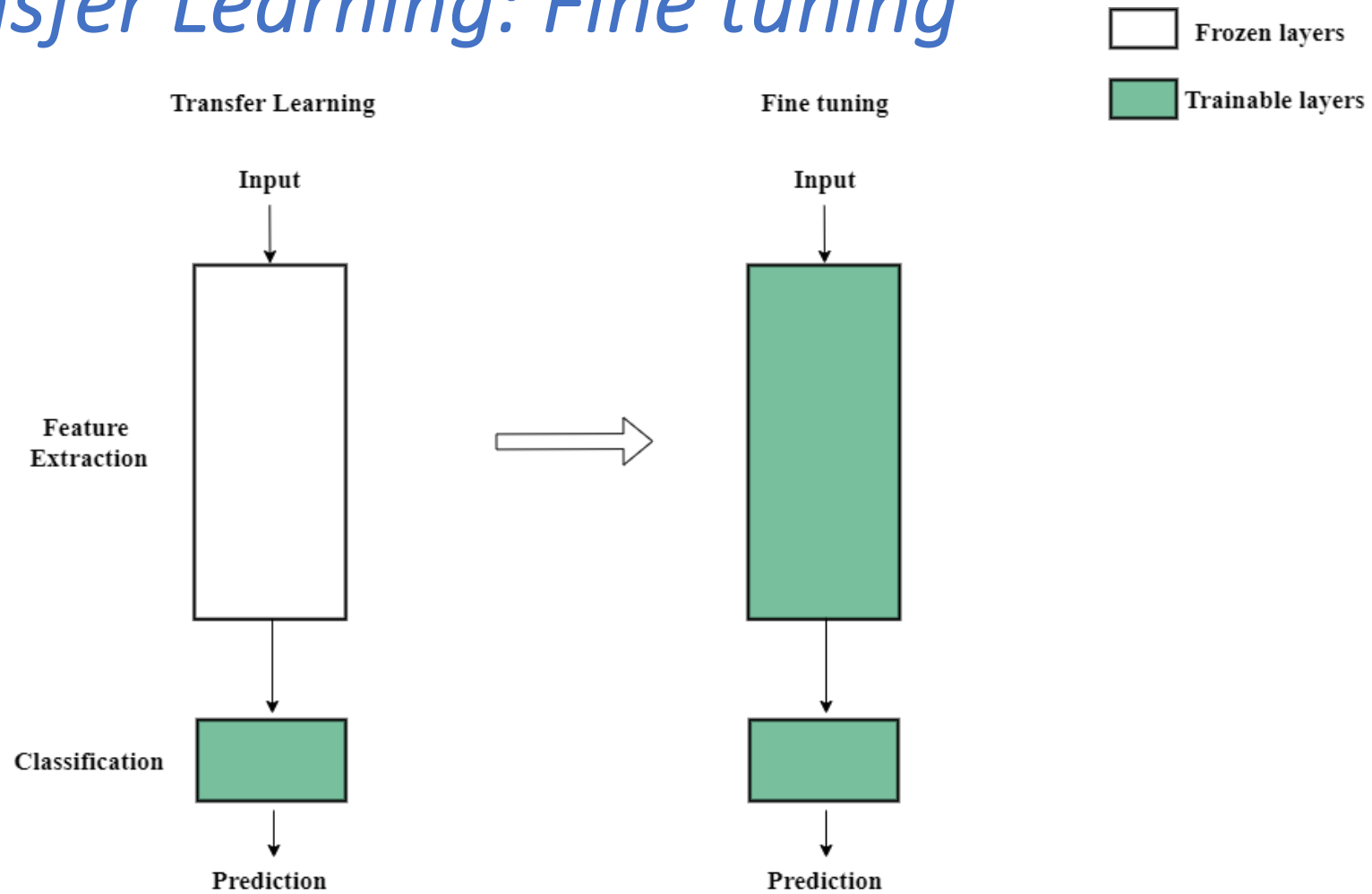
Transfer Learning is widely used for several reasons:

- **Significantly reduces training time** (fast convergence, freezing earlier layers).
- **Prevents overfitting** when working with small datasets.
- **Achieves impressive performance** compared to training a model from scratch.

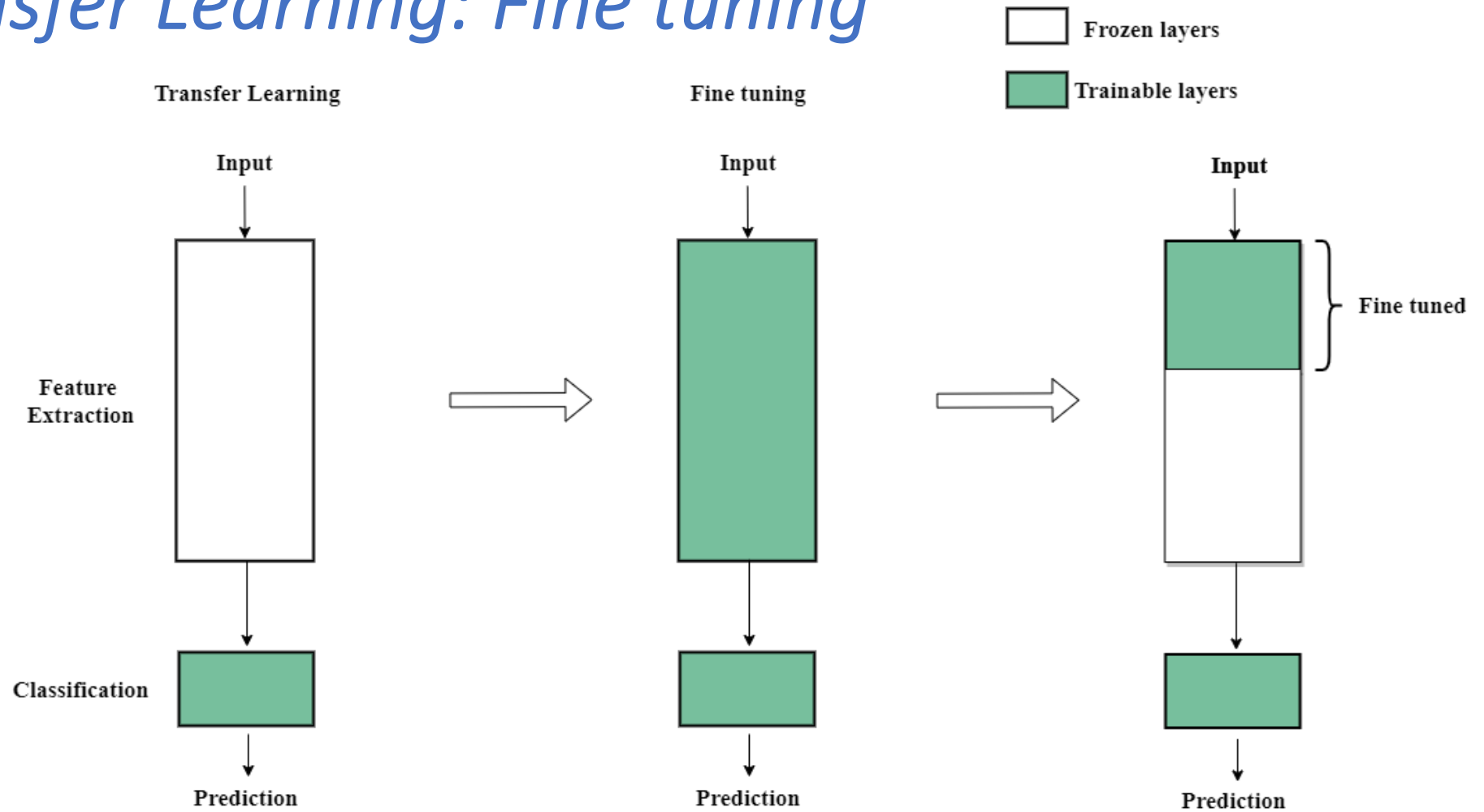
Why Transfer Learning?

	Training from scratch	Transfer Learning
Dataset size	Large	Small
Model complexity	High	Low
Training time	Long	Short
Generalization	Weak	Strong

Transfer Learning: Fine tuning



Transfer Learning: Fine tuning



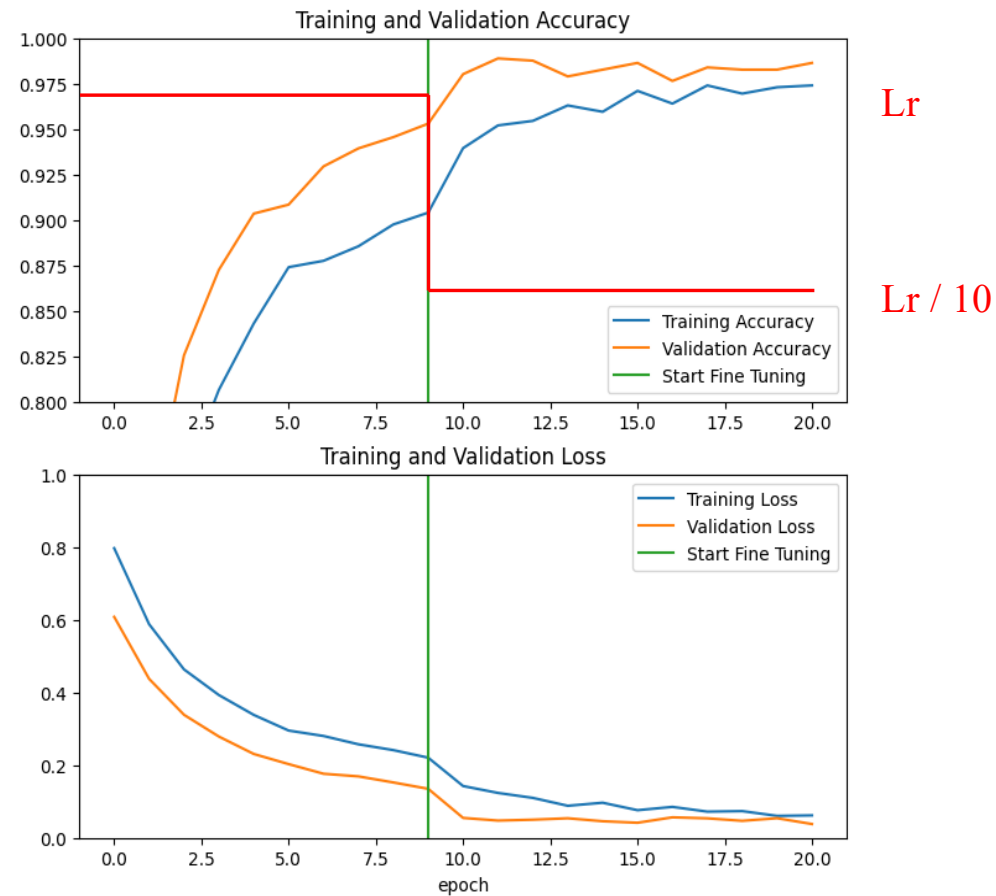
Transfer Learning: Fine tuning

Fine-tuning is a step that follows ***Transfer Learning***. The difference is that, in the case of *fine-tuning*, all the layers of the model are trained with a small learning rate.

Fine-tuning provides greater flexibility in adapting a model to a new dataset, which can result in better performance.

However, models are more prone to overfitting with *fine-tuning* compared to *Transfer Learning*.

Transfer Learning: Fine tuning

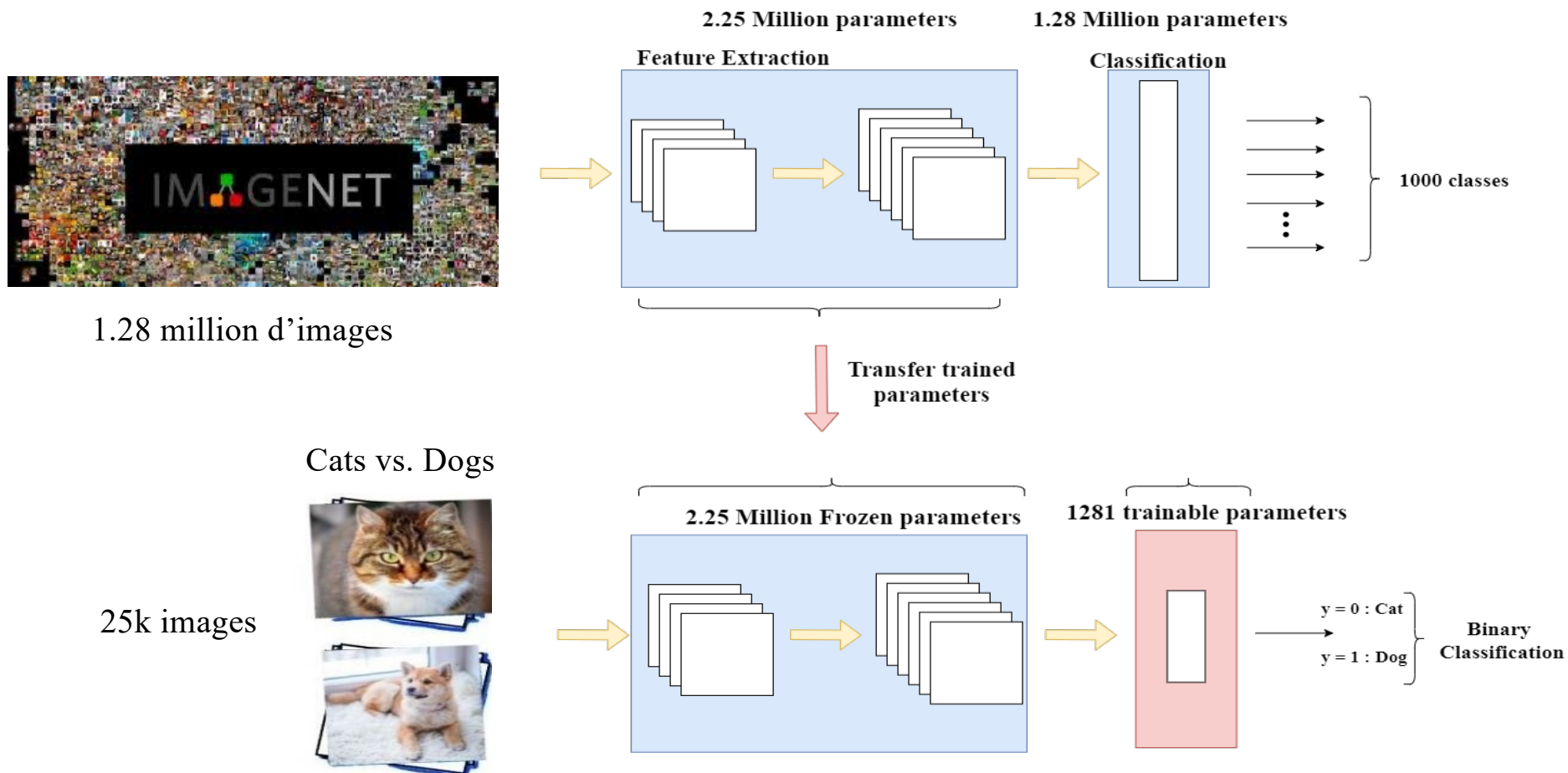


Ref.: https://www.tensorflow.org/tutorials/images/transfer_learning

Transfer Learning with or without Fine tuning?

Cases	Transfer Learning & Fine Tuning	Fine Tuning	Training from scratch
Small dataset and similar to the original one	✓		
Large dataset and similar to the original one	✓	✓	
Large dataset and different from the original one			✓

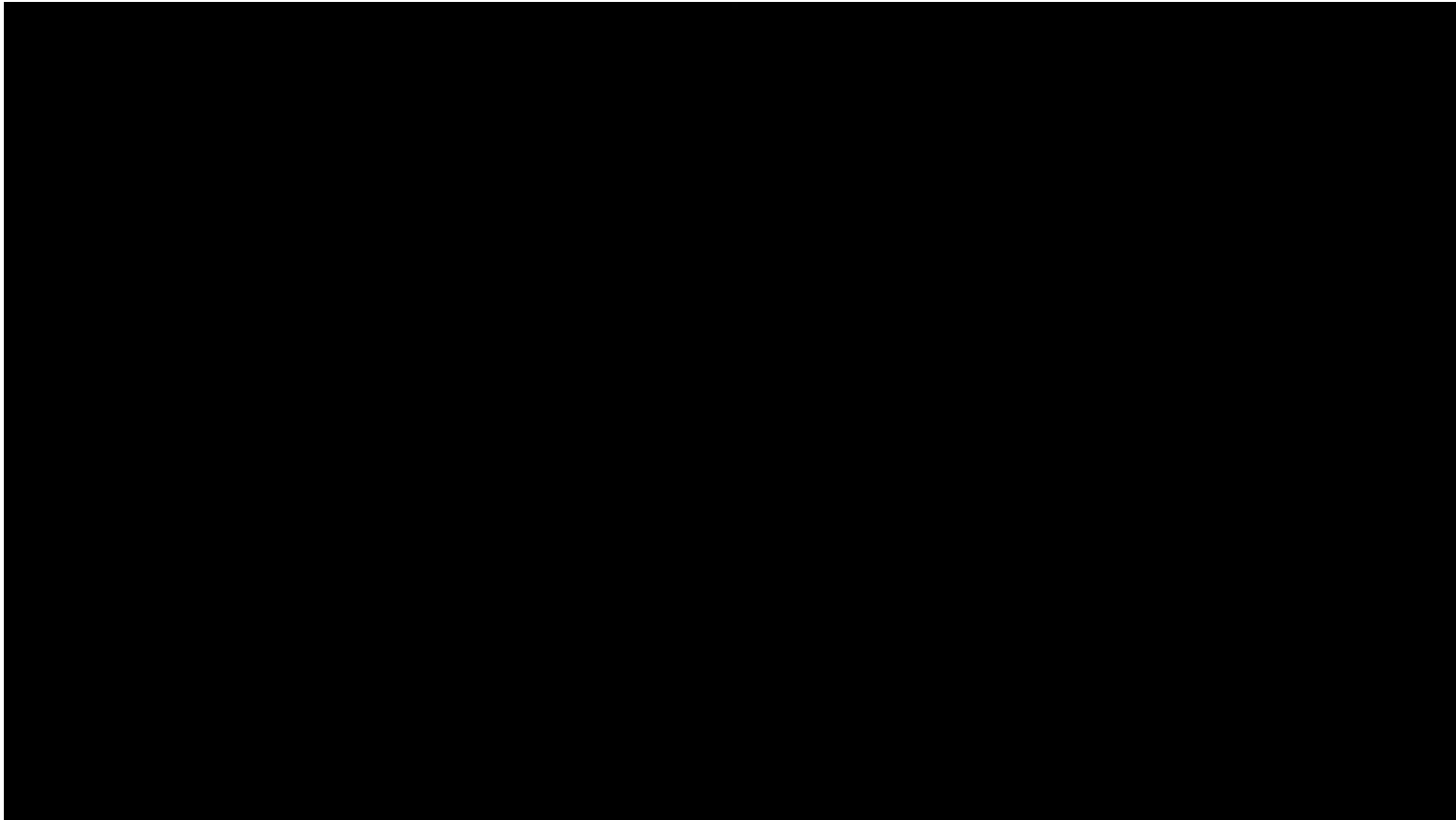
Transfer Learning with MobileNet V2



[1] SANDLER, Mark et al. Mobilenetv2: CVPR. 2018. p. 4510-4520.

[2] https://www.tensorflow.org/tutorials/images/transfer_learning

Transfer Learning & Fine Tuning



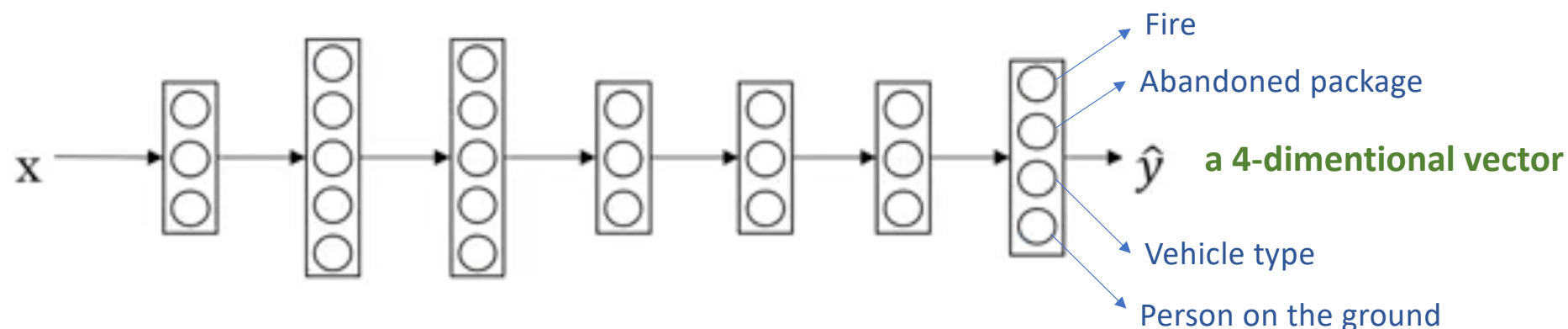
Transfer Learning vs. Multi-task Learning

An application like scene analysis for video surveillance:

- Detect fire:
Detect flames or smoke in an image.
- Detect abandoned package
Identify objects like bags or suitcases left unattended.
- Vehicle type classification
Differentiating between cars and trucks.
- Detect a person lying on the ground
Detect people in unusual postures, especially lying down, which could indicate medical emergencies or accidents.

Transfer Learning vs. Multi-task Learning

Network architecture



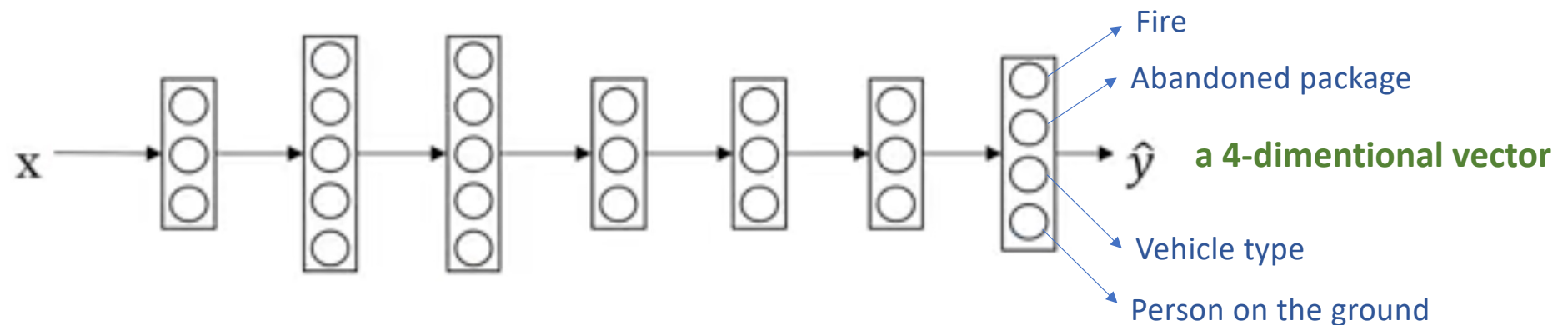
$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y})$ and $\hat{\mathbf{y}}$ is (4,1) vector.

$$J = \frac{1}{m} \sum_{i=1}^m \sum_{j=1}^4 \mathcal{L}(\hat{y}_j^{(i)}, y_j^{(i)})$$

$$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) = -[y \log(\hat{\mathbf{y}}) + (1 - y) \log(1 - \hat{\mathbf{y}})]$$

Transfer Learning vs. Multi-task Learning

Network architecture



The advantage:

It makes use of all of the train dataset (for all tasks) to build the NN and to train it. Even partially labeled example can be used.

Transfer Learning vs. Multi-task Learning

In multitask learning:

- Training on a set of tasks could benefit from having shared lower-level features.
- Train a sufficient large NN to handle all of the tasks.
- Amount of training data for each task: A balanced amount is not very important.
- It is much more effective than training a separate NN for each task.